DSA0603 Data Handling and Visualization

Slot B

CO1 – Assessment Test 3 (AT3)

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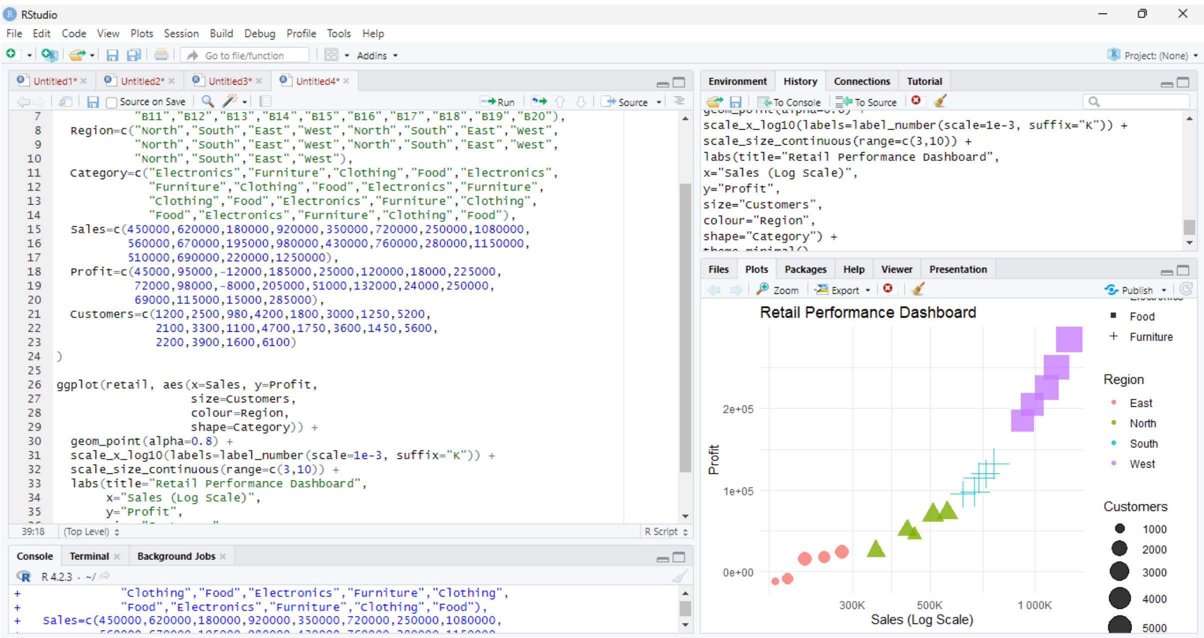
Question 1 – Integrated Retail Visualization

```
library(ggplot2)
library(scales)
retail <- data.frame(
  Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",
         "B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
  Region=c("North","South","East","West","North","South","East","West",
         "North","South","East","West","North","South","East","West",
         "North","South","East","West"),
  Category=c("Electronics","Furniture","Clothing","Food","Electronics",
           "Furniture","Clothing","Food","Electronics","Furniture",
           "Clothing","Food","Electronics","Furniture","Clothing",
           "Food","Electronics","Furniture","Clothing","Food"),
  Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
        560000,670000,195000,980000,430000,760000,280000,1150000,
        510000,690000,220000,1250000),
  Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
         72000,98000,-8000,205000,51000,132000,24000,250000,
         69000,115000,15000,285000),
  Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
            2100,3300,1100,4700,1750,3600,1450,5600,
            2200,3900,1600,6100)
)
```

```
ggplot(retail, aes(x=Sales, y=Profit,
                   size=Customers,
                   colour=Region,
                   shape=Category)) +
  geom_point(alpha=0.8) +
  scale_x_log10(labels=label_number(scale=1e-3, suffix="K")) +
  scale_size_continuous(range=c(3,10)) +
  labs(title="Retail Performance Dashboard",
       x="Sales (Log Scale)",
       y="Profit",
       size="Customers",
       colour="Region",
       shape="Category") +
  theme_minimal()
```



Interpretation and Justification

- **X-axis = Sales** and **Y-axis = Profit** show revenue and profitability simultaneously.
- **Point size = Customers** represents customer base magnitude.

- **Colour = Region** distinguishes geographical performance.
- **Shape = Category** differentiates product groups.

Scale Transformation

- Sales values range from **180,000 to 1,250,000**, so a **log10 scale** reduces skewness and prevents high-sales branches from dominating the plot.

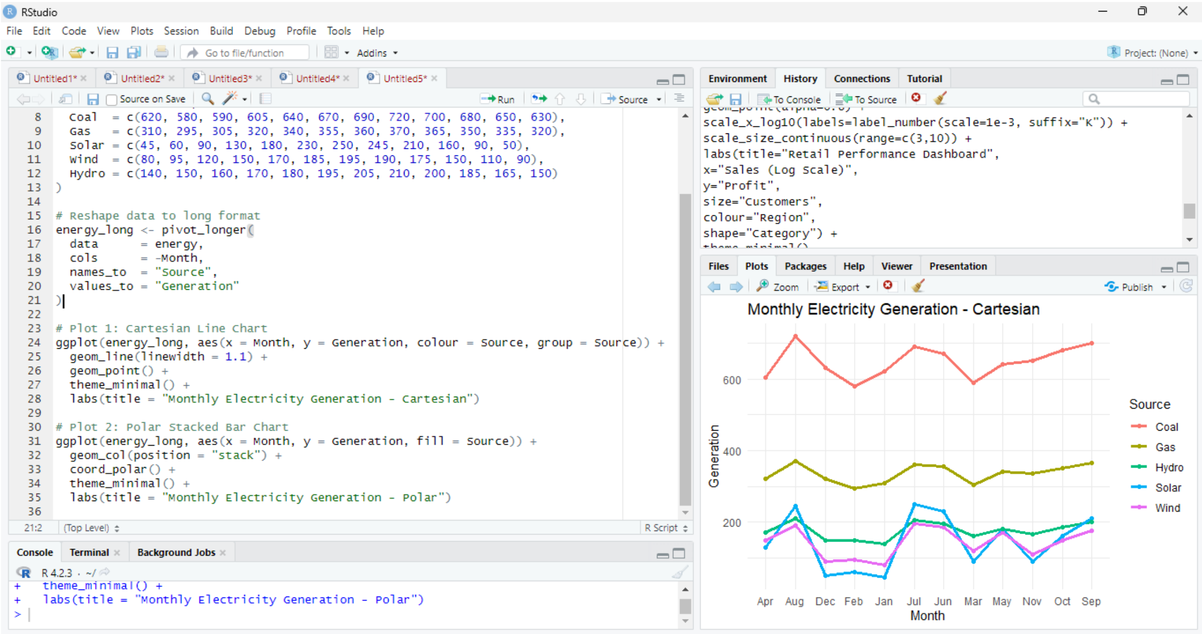
Key Insights

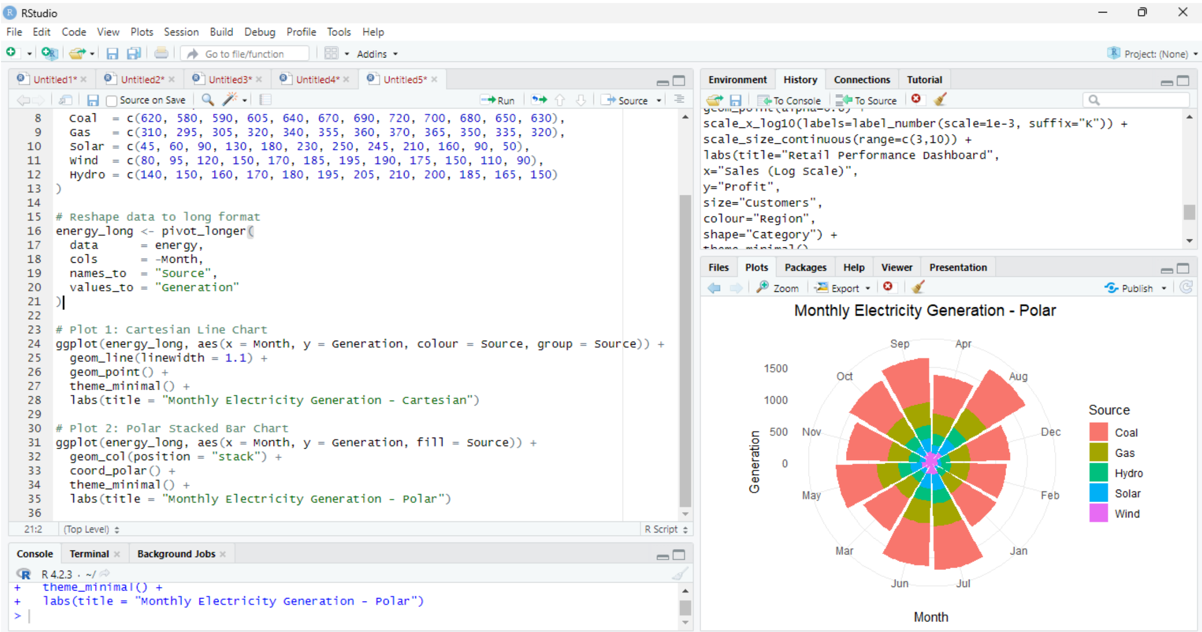
- West region branches generally achieve the highest sales and profits.
- Clothing category includes negative-profit branches, indicating weaker performance.
- Larger customer bases are associated with higher sales in most regions.

Question 2 – Cartesian vs Polar Coordinates

```
library(ggplot2)

library(tidyr)

# Create the data frame

energy <- data.frame(

  Month = month.abb,

  Coal  = c(620, 580, 590, 605, 640, 670, 690, 720, 700, 680, 650, 630),

  Gas   = c(310, 295, 305, 320, 340, 355, 360, 370, 365, 350, 335, 320),

  Solar = c(45, 60, 90, 130, 180, 230, 250, 245, 210, 160, 90, 50),

  Wind  = c(80, 95, 120, 150, 170, 185, 195, 190, 175, 150, 110, 90),

  Hydro = c(140, 150, 160, 170, 180, 195, 205, 210, 200, 185, 165, 150)

)


# Reshape data to long format

energy_long <- pivot_longer(

  data      = energy,

  cols      = -Month,
```

```r
  names_to  = "Source",
  values_to = "Generation"
)

# Plot 1: Cartesian Line Chart
ggplot(energy_long, aes(x = Month, y = Generation, colour = Source, group = Source)) +
  geom_line(linewidth = 1.1) +
  geom_point() +
  theme_minimal() +
  labs(title = "Monthly Electricity Generation - Cartesian")

# Plot 2: Polar Stacked Bar Chart
ggplot(energy_long, aes(x = Month, y = Generation, fill = Source)) +
  geom_col(position = "stack") +
  coord_polar() +
  theme_minimal() +
  labs(title = "Monthly Electricity Generation - Polar")
```





Comparison

Cartesian Coordinates

- Best for comparing exact monthly values.
- Clearly shows trends, peaks, and seasonal changes.

Polar Coordinates

- Emphasizes cyclical annual patterns.
- Useful for communicating seasonality but not precise comparisons.

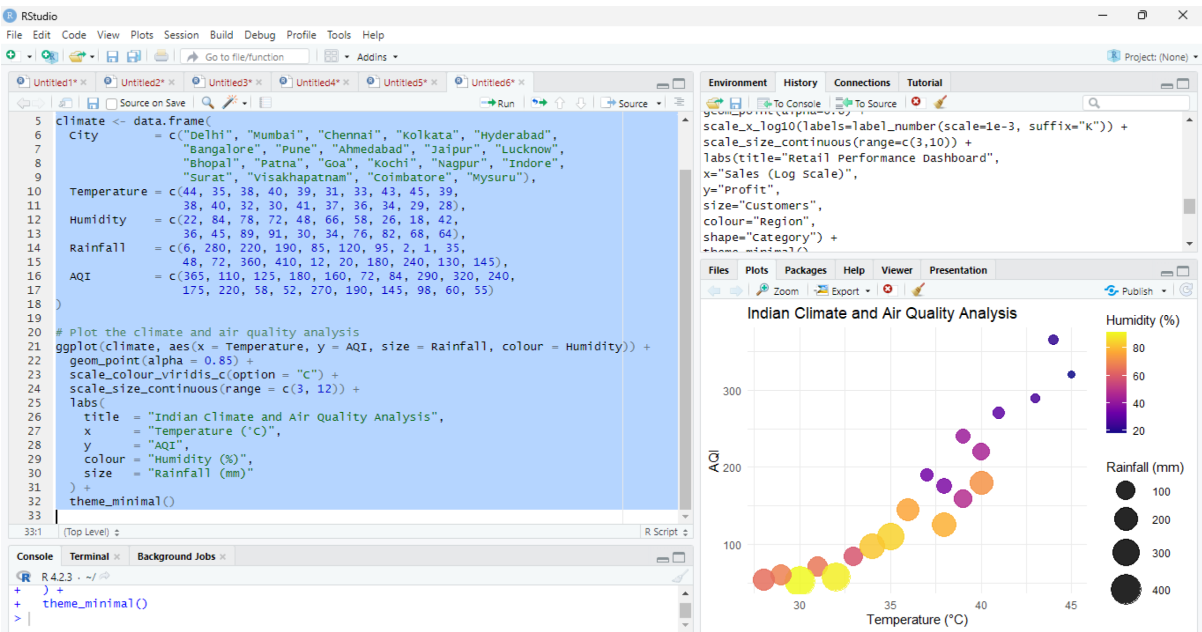
Better Analytical Insight

Cartesian coordinates provide better analytical insight because line charts preserve a common baseline and support accurate month-to-month comparison

Question 3 – Climate Visualization with Colour Scales

```
library(ggplot2)

library(scales)


# Create the climate data frame

climate <- data.frame(

  City        = c("Delhi", "Mumbai", "Chennai", "Kolkata", "Hyderabad",

                  "Bangalore", "Pune", "Ahmedabad", "Jaipur", "Lucknow",

                  "Bhopal", "Patna", "Goa", "Kochi", "Nagpur", "Indore",

                  "Surat", "Visakhapatnam", "Coimbatore", "Mysuru"),

  Temperature = c(44, 35, 38, 40, 39, 31, 33, 43, 45, 39,

                  38, 40, 32, 30, 41, 37, 36, 34, 29, 28),

  Humidity    = c(22, 84, 78, 72, 48, 66, 58, 26, 18, 42,

                  36, 45, 89, 91, 30, 34, 76, 82, 68, 64),

  Rainfall    = c(6, 280, 220, 190, 85, 120, 95, 2, 1, 35,

                  48, 72, 360, 410, 12, 20, 180, 240, 130, 145),

  AQI         = c(365, 110, 125, 180, 160, 72, 84, 290, 320, 240,

                  175, 220, 58, 52, 270, 190, 145, 98, 60, 55)

)
```

```
# Plot the climate and air quality analysis

ggplot(climate, aes(x = Temperature, y = AQI, size = Rainfall, colour =
Humidity)) +

  geom_point(alpha = 0.85) +

  scale_colour_viridis_c(option = "C") +

  scale_size_continuous(range = c(3, 12)) +

  labs(

    title  = "Indian Climate and Air Quality Analysis",

    x      = "Temperature (°C)",

    y      = "AQI",

    colour = "Humidity (%)",

    size   = "Rainfall (mm)"

  ) +

  theme_minimal()
```



Design Justification

- **Temperature** on x-axis and **AQI** on y-axis reveal heat–pollution relationships.
- **Rainfall** encoded by point size highlights monsoon intensity.
- **Humidity** encoded by colour uses the **Viridis palette**, which is perceptually uniform and colour-blind friendly.

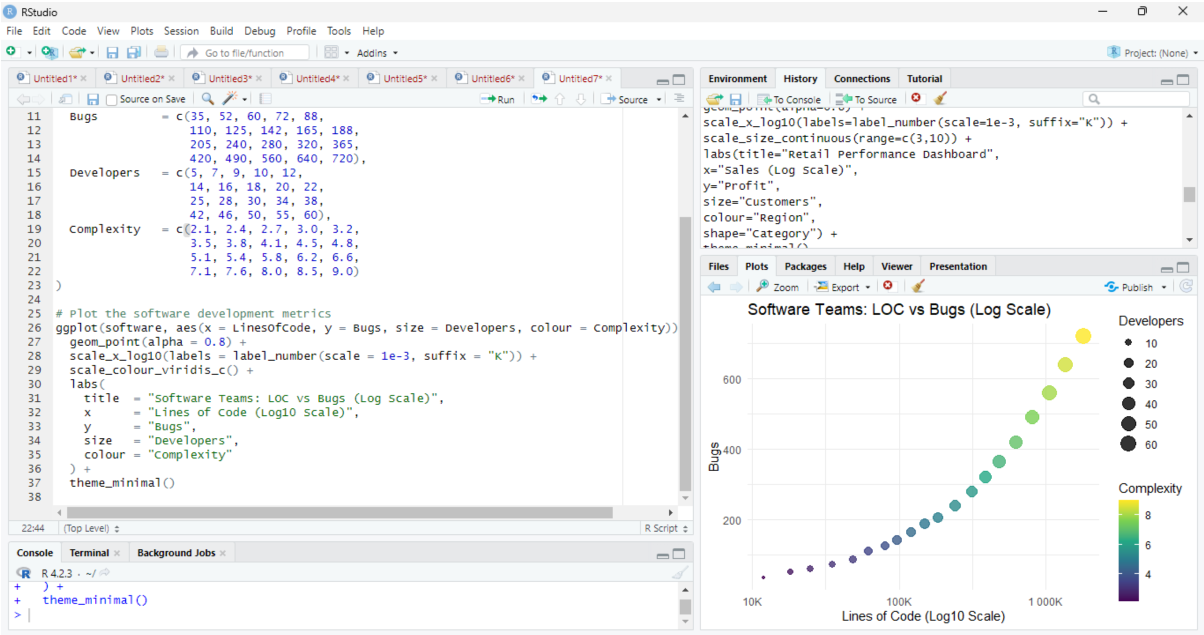
Interpretation

- Delhi and Jaipur show very high AQI and temperature.
- Kochi and Goa have high rainfall and humidity but low AQI.
- Cooler southern cities generally exhibit better air quality.

Question 4 – Nonlinear Transformation for Skewed Software Data

```
library(ggplot2)

library(scales)


# Create the software team data frame

software <- data.frame(

  Team         = paste("T", 1:20, sep = ""),

  LinesOfCode  = c(12000, 18000, 25000, 35000, 48000,

                62000, 81000, 96000, 120000, 150000,

                185000, 240000, 310000, 390000, 480000,

                620000, 810000, 1050000, 1350000, 1800000),

  Bugs         = c(35, 52, 60, 72, 88,

                110, 125, 142, 165, 188,

                205, 240, 280, 320, 365,

                420, 490, 560, 640, 720),

  Developers   = c(5, 7, 9, 10, 12,

                14, 16, 18, 20, 22,
```

```
                25, 28, 30, 34, 38,
                42, 46, 50, 55, 60),
  Complexity   = c(2.1, 2.4, 2.7, 3.0, 3.2,
                3.5, 3.8, 4.1, 4.5, 4.8,
                5.1, 5.4, 5.8, 6.2, 6.6,
                7.1, 7.6, 8.0, 8.5, 9.0)
)

# Plot the software development metrics
ggplot(software, aes(x = LinesOfCode, y = Bugs, size = Developers, colour = Complexity)) +
  geom_point(alpha = 0.8) +
  scale_x_log10(labels = label_number(scale = 1e-3, suffix = "K")) +
  scale_colour_viridis_c() +
  labs(
    title  = "Software Teams: LOC vs Bugs (Log Scale)",
    x      = "Lines of Code (Log10 Scale)",
    y      = "Bugs",
    size   = "Developers",
    colour = "Complexity"
  ) +
  theme_minimal()
```



Justification

- Lines of Code ranges from **12,000 to 1,800,000**, creating severe right skew.
- A **log10 transformation** compresses extreme values and spreads smaller values.

Improvement over Linear Scale

- Reveals relationships among small and medium teams that would otherwise cluster together.
- Makes growth patterns easier to compare.
- Improves readability and reduces visual dominance of very large projects.

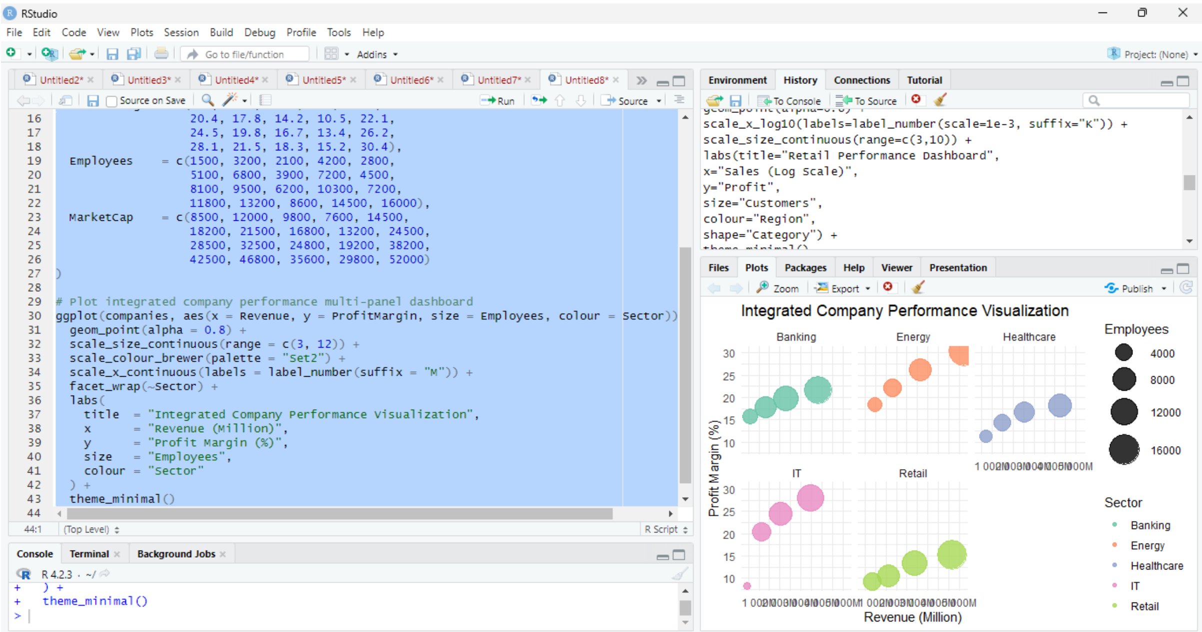
Question 5 – Integrated Financial Analytics Visualization

library(ggplot2)

library(scales)

```r
# Create the company metrics data frame
companies <- data.frame(
  Company      = paste("C", 1:20, sep = ""),
  Sector       = c("IT", "Banking", "Healthcare", "Retail", "Energy",
                   "IT", "Banking", "Healthcare", "Retail", "Energy",
                   "IT", "Banking", "Healthcare", "Retail", "Energy",
                   "IT", "Banking", "Healthcare", "Retail", "Energy"),
  Revenue      = c(420, 580, 720, 850, 980,
                   1150, 1320, 1480, 1650, 1820,
                   2050, 2280, 2550, 2840, 3150,
                   3480, 3860, 4250, 4680, 5200),
  ProfitMargin = c(8.2, 15.6, 11.4, 9.3, 18.5,
                   20.4, 17.8, 14.2, 10.5, 22.1,
                   24.5, 19.8, 16.7, 13.4, 26.2,
                   28.1, 21.5, 18.3, 15.2, 30.4),
  Employees    = c(1500, 3200, 2100, 4200, 2800,
                   5100, 6800, 3900, 7200, 4500,
                   8100, 9500, 6200, 10300, 7200,
                   11800, 13200, 8600, 14500, 16000),
  MarketCap    = c(8500, 12000, 9800, 7600, 14500,
                   18200, 21500, 16800, 13200, 24500,
                   28500, 32500, 24800, 19200, 38200,
                   42500, 46800, 35600, 29800, 52000)
)

# Plot integrated company performance multi-panel dashboard
ggplot(companies, aes(x = Revenue, y = ProfitMargin, size = Employees, colour = Sector)) +
  geom_point(alpha = 0.8) +
  scale_size_continuous(range = c(3, 12)) +
```

```
scale_colour_brewer(palette = "Set2") +
scale_x_continuous(labels = label_number(suffix = "M")) +
facet_wrap(~Sector) +
labs(
  title  = "Integrated Company Performance Visualization",
  x      = "Revenue (Million)",
  y      = "Profit Margin (%)",
  size   = "Employees",
  colour = "Sector"
) +
theme_minimal()
```



Design Justification

- **Revenue** on x-axis and **Profit Margin** on y-axis capture scale and profitability.
- **Employees** mapped to size indicate organisational scale.
- **Sector** mapped to colour and faceting avoids overlap and improves comparison.
- **Set2 palette** provides balanced categorical colours with reduced perceptual bias.

Interpretation

- Energy companies show the highest profit margins.
- IT companies exhibit strong profitability with large revenues.
- Retail companies have larger employee counts but comparatively lower margins.
- Banking companies occupy a middle performance range.

Conclusion

The integrated bubble-facet visualization communicates five variables simultaneously while maintaining clarity, reducing clutter, and supporting sector-wise comparison for strategic financial analysis.